Yunsoo Ha

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Academic Appointments

- **Cornell University** — *2025 – Present*
  Postdoctoral Associate, School of Civil and Environmental Engineering — USA
- **National Laboratory of the Rockies** — *2024 – 2025*
  Postdoctoral Researcher, Computational Science Center — USA

Education

- **Ph.D. in Industrial and Systems Engineering** — *2018 – 2024*
  North Carolina State University — USA
  - Advisor: Sara Shashaani
  - Committee: Yunan Liu, Reha Uzsoy, Quoc Tran-Dinh, Naihuan Jing
  - Dissertation: *Expediting Stochastic Derivative-Free Optimization*
  - *ISE Distinguished Dissertation Award of the 2024 CA Anderson Awards*
  - *Second Place Winner of the 2025 Pritsker Doctoral Dissertation Award by IISE*
- **Master of Operations Research** — *2018 – 2021*
  North Carolina State University — USA
- **M.S. in Logistics, School of Air Transport, Transportation, and Logistics** — *2015 – 2017*
  Korea Aerospace University — South Korea
- **B.S. in Logistics, School of Air Transport, Transportation, and Logistics** — *2010 – 2015*
  Korea Aerospace University — South Korea

Research Interests

Methods: Simulation Optimization, Stochastic and Derivative-Free Optimization, Monte Carlo Methods
Directions: Decision-Focused Learning, Digital Twin Optimization, Quantum-Classical Optimization

Publications

Peer-Reviewed Journal Articles

[1] **Adaptive Sampling Bi-Fidelity Stochastic Trust Region Method for Derivative-Free Stochastic Optimization**.
Yunsoo Ha, Juliane Mueller.
Mathematical Programming Computation (2026).

[2] **Complexity of Zeroth- and First-Order Stochastic Trust-Region Algorithms**.
Yunsoo Ha, Raghu Pasupathy, Sara Shashaani.
SIAM Journal on Optimization (2025).

[3] **Two-Stage Estimation and Variance Modeling for Latency-Constrained Variational Quantum Algorithms**.
Yunsoo Ha, Sara Shashaani, Matt Menickelly.
INFORMS Journal on Computing (2025).

[4] **Iteration Complexity and Finite-Time Efficiency of Adaptive Sampling Trust-Region Methods for Stochastic Derivative-Free Optimization**.
Yunsoo Ha, Sara Shashaani.
IISE Transactions (2025).

[5] **Latency Considerations for Stochastic Optimizers in Variational Quantum Algorithms**.
Matt Menickelly, **Yunsoo Ha**, Matthew Otten.
Quantum (2023).

[6] **A Decision Model to Determine the Number of Shuttles in a Tier-to-Tier SBS/RS**.
Yunsoo Ha, Junjae Chae.
International Journal of Production Research (2019).

[7] **Free Balancing for a Shuttle-Based Storage and Retrieval System**.
Yunsoo Ha, Junjae Chae.
Simulation Modelling Practice and Theory (2018).

Manuscripts Under Review

[1] **Computing Equilibria in Simulation-Based Insurance Markets with Discontinuous Demand**.
Yunsoo Ha, Linda Nozick.

[2] **Adaptive Regularization within Trust Region Methods for Stochastic Nonconvex Optimization**.
Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.
🏆 *Third Place Poster Award, 2026 INFORMS Simulation Society Research Workshop*

Working Papers

[1] **Multilevel Monte Carlo Trust-Region Method with Adaptive Sampling for Nested Simulation Optimization**.
Yunsoo Ha, Hyunwoo Shin.

[2] **Stochastic Derivative-Free Optimization with Adaptive Regularization and Adaptive Sampling Trust Region Methods**.
Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.

[3] **Decision-Focused Input Modeling for Queueing Systems**.
Yunsoo Ha.

Peer-Reviewed Conference Proceedings

[1] **Multi-Fidelity Stochastic Trust Region Method with Adaptive Sampling**.
Yunsoo Ha, Juliane Mueller.
Winter Simulation Conference 2025.

[2] **Towards Greener Stochastic Derivative-Free Optimization with Trust Regions and Adaptive Sampling**.
Yunsoo Ha, Sara Shashaani.
Winter Simulation Conference 2023.

[3] **Improved Complexity of Trust-Region Optimization for Zeroth-Order Stochastic Oracles With Adaptive Sampling**.
Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.
Winter Simulation Conference 2021.

Honors and Awards

- **Third Place Poster Presentation Award** *2026*
  INFORMS Simulation Society Research Workshop
- **Second Place Winner of the 2025 Pritsker Doctoral Dissertation Award** *2025*
  Institute of Industrial and Systems Engineers
- **Outstanding Reviewer Award** *2024*
  Winter Simulation Conference
- **ISE Distinguished Dissertation Award of the 2024 C.A. Anderson Awards** *2024*
  North Carolina State University
- **Travel Award for the Annual Midwest Optimization Meeting** *2023*
  Michigan State University
- **Mentored Teaching Fellowship** *2022*
  North Carolina State University
- **Scholarship for Excellent Academic Records** *2010, 2013–2016*
  Korea Aerospace University

Teaching Experience

- **ISE 362: Stochastic Models in Industrial Engineering** *Spring 2023*
  Instructor North Carolina State University
- **ISE 441: Introduction to Simulation** *Fall 2019, Spring 2020, Fall 2020, Fall 2022*
  Teaching Assistant North Carolina State University
- **ISE 748: Quality Engineering** *Spring 2019*
  Teaching Assistant North Carolina State University
- **ISE 498: Senior Design Project** *Fall 2018*
  Teaching Assistant North Carolina State University
- **Analysis of Logistics System** *Fall 2015, Fall 2016*
  Teaching Assistant Korea Aerospace University

Research Mentoring

- **Graduate Students**
  - Qian Ni (Ph.D. Student, Civil and Environmental Engineering, Cornell University)
  - Hyunwoo Shin (Ph.D. Candidate, Industrial and Systems Engineering, Virginia Tech; NREL Summer Intern)
  - Hyung Khee Eun (Ph.D. Candidate, Industrial and Systems Engineering, North Carolina State University)
- **Undergraduate Students**
  - Kevin Xu (Statistics, North Carolina State University)
  - Wes Hankinson (Industrial and Systems Engineering, North Carolina State University)

Invited Talks

- **INFORMS Annual Meeting** *2026*
  Computing Equilibria in Simulation-Based Insurance Markets with Discontinuous Demand
- **INFORMS Optimization Society Conference** *2026*
  A Derivative-Free Solver for Multi-Region Stochastic Equilibrium Problems
- **INFORMS Virginia Tech Student Chapter Seminar Series** *2025*
  Expediting Simulation Optimization
- **INFORMS Annual Meeting** *2025*
  Adaptive Sampling-Based Bi-Fidelity Stochastic Trust Region Method for Derivative-Free Stochastic Optimization
- **International Symposium on Mathematical Programming** *2024*
  Two talks: Multi-Fidelity Stochastic Trust-Region Method; Is Building First-Order Simulation Oracles Really Worth It?
- **INFORMS Optimization Society Conference** *2024*
  First-Order Trust-Region Methods with Adaptive Sampling
- **Annual Midwest Optimization Meeting** *2023*
  Common Random Numbers and Complexity in Simulation Optimization with Adaptive Sampling
- **INFORMS Annual Meeting** *2023*
  Consistency and Complexity of Adaptive Sampling-Based Trust-Region Optimization
- **Modeling and Optimization: Theory and Applications (MOPTA)** *2023*
  Expediting Stochastic Derivative-Free Optimization
- **INFORMS Annual Meeting** *2022*
  Complexity Analysis of Trust-Region Optimization with Adaptive Sampling for Zeroth-Order Stochastic Oracles
- **INFORMS Optimization Society Conference** *2022*
  Adaptive Sampling Trust-Region Optimization with Diagonal Hessian for Derivative-Free Stochastic Oracles

Conference Presentations

- **INFORMS Simulation Society Research Workshop** *2026*
  Adaptive Regularization within Trust Region Methods for Stochastic Nonconvex Optimization (Poster)
- **Winter Simulation Conference** *2025*
  Multi-Fidelity Stochastic Trust Region Method with Adaptive Sampling
- **Institute of Industrial and Systems Engineers Annual Conference** *2025*
  First-Order Trust-Region Methods with Adaptive Sampling
- **Winter Simulation Conference** *2023*
  Towards Greener Stochastic Derivative-Free Optimization with Trust Regions and Adaptive Sampling
- **Winter Simulation Conference** *2021*
  Improved Complexity of Trust-Region Optimization for Zeroth-Order Stochastic Oracles with Adaptive Sampling
- **Winter Simulation Conference** *2020*
  Traffic Signal Control Simulation and Optimization (Poster)

Academic Service

- **Journal Refereeing**
  - Computational Optimization and Applications
  - Journal of Scientific Computing
  - Journal of Simulation
  - Mathematical Programming
  - Mathematical Programming Computation
  - Numerical Algorithms
  - Simulation Modelling Practice and Theory
  - Statistics and Computing
- **Conference Refereeing**
  - Winter Simulation Conference, 2024–2026
- **Session Chair**
  - INFORMS Annual Meeting, 2022, 2023, 2025
  - Modeling and Optimization: Theory and Applications (MOPTA), 2023
  - Winter Simulation Conference, 2023